

Question 1

Not yet answered

Marked out of 2.00

 Flag question

For low-pass RL filter with $R = 20 \ \Omega$ and $L = 2 \text{ mH}$, the corner frequency is:

Select one:

- ☐ a. 796 Hz
- ☐ b. 19,894 Hz
- ☐ c. 3,979 Hz
- ☐ d. 1,592 Hz
- ☐ e. 7,958 Hz

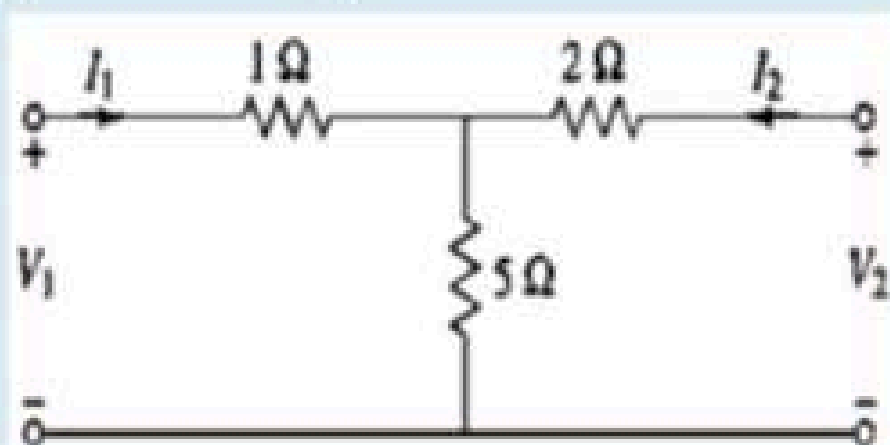
Question 2

Not yet answered

Marked out of 2.00

Flag question

For the circuit below, the transmission parameter (B) values is



Select one:

- ☐ a. $4/5$
- ☐ b. $1/5$
- ☐ c. $6/5$
- ☐ d. $7/5$
- ☐ e. $17/5$

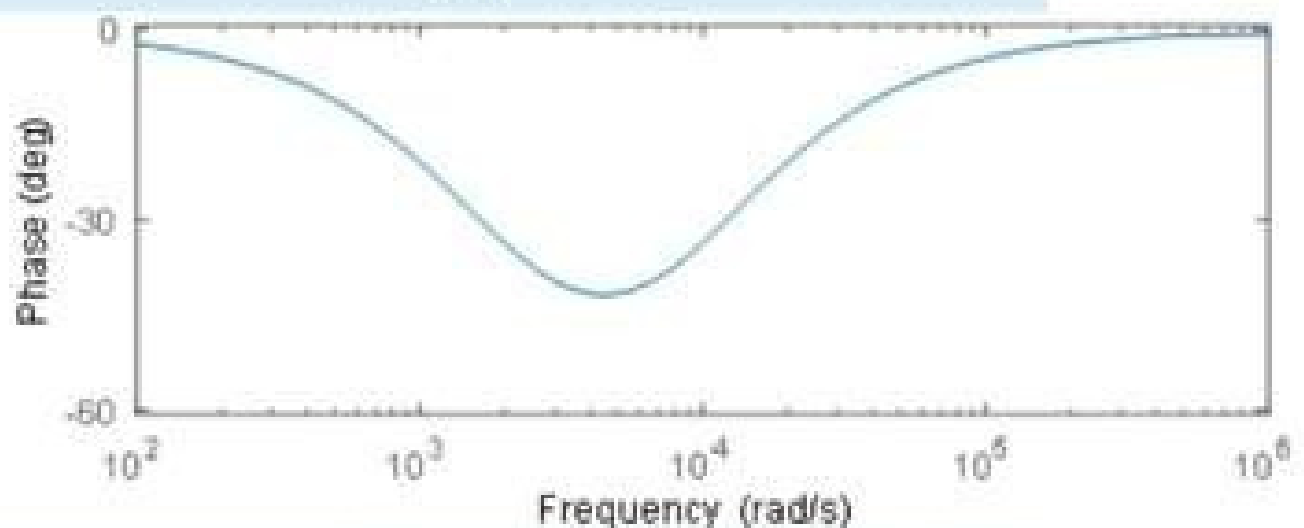
Question 3

Not yet answered

Marked out of 2.00

Flag question

The phase Bode diagram for the figure below is for transfer function $H(s) =$:



Select one:

- ☐ a. $(0.0001 S + 1) / (0.0005 S + 1)$
- ☐ b. $(0.1 S + 1) / (0.5 S + 1)$
- ☐ c. $(0.01 S + 1) / (0.05 S + 1)$
- ☐ d. $(0.001 S + 1) / (0.005 S + 1)$

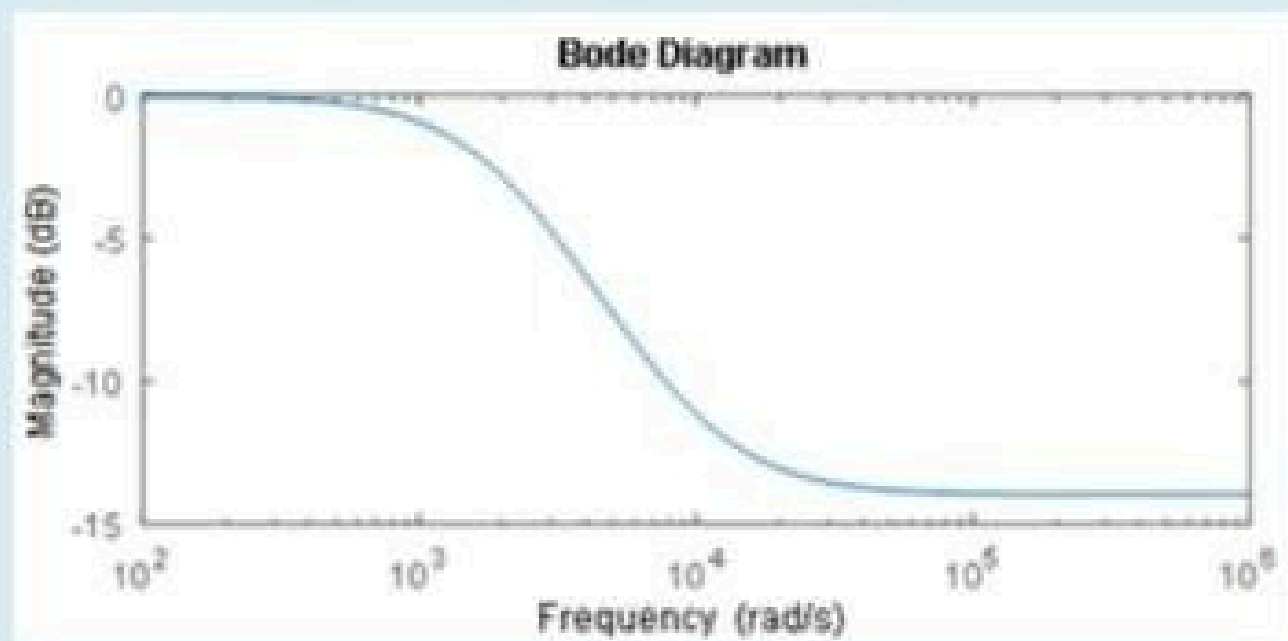
Question 4

Not yet answered

Marked out of 2.00

Flag question

The magnitude Bode diagram for the figure below is for transfer function $H(s) =$:



Select one:

- ☐ a. $(0.01 S + 1) / (0.05 S + 1)$
- ☐ b. $(0.0001 S + 1) / (0.0005 S + 1)$
- ☐ c. $(0.001 S + 1) / (0.005 S + 1)$
- ☐ d. $(0.1 S + 1) / (0.5 S + 1)$

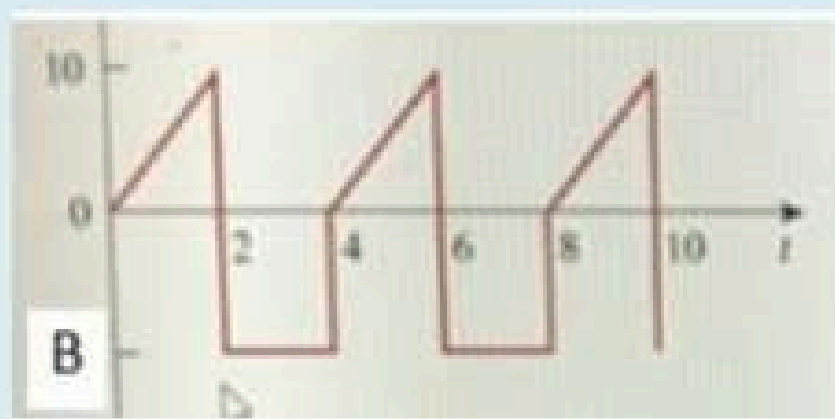
Question 5

Not yet answered

Marked out of 2.00

Flag question

Find the rms value for the signal below if $B = -9$.



Select one:

- ☐ a. 6.42
- ☐ b. 8.16
- ☐ c. 5.89
- ☐ d. 6.98
- ☐ e. 7.56

Question 6

Not yet answered

Marked out of 2.00

Flag question

The time-domain voltages in Y-connected load is given as:

$V_{an} = 400 \cos(\omega t + 30^\circ)$, $V_{bn} = 400 \cos(\omega t - 90^\circ)$, $V_{cn} = 400 \cos(\omega t + 150^\circ)$. Then V_{ab} is

Select one:

- ☐ a. $381 \cos(\omega t + 60^\circ)$
- ☐ b. $693 \cos(\omega t + 60^\circ)$
- ☐ c. $381 \cos(\omega t + 180^\circ)$
- ☐ d. $693 \cos(\omega t - 60^\circ)$
- ☐ e. $381 \cos(\omega t - 60^\circ)$

Question 7

Not yet answered

Marked out of 2.00

 Flag question

A 25-kVA ideal power transformer has 500 turns on the primary and 50 turns on the secondary winding. The primary is connected to 2500 V supply, Find load current in the secondary side.

Select one:

- ☐ a. 83.3 A
- ☐ b. 200 A
- ☐ c. 50 A
- ☐ d. 250 A
- ☐ e. 100 A

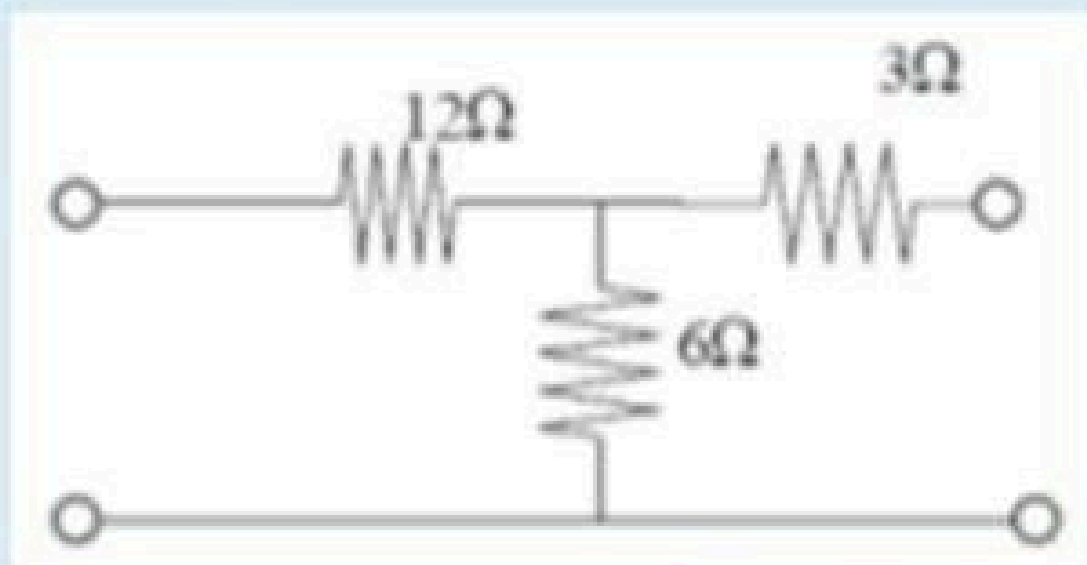
Question 8

Not yet answered

Marked out of 1.00

Flag question

For the circuit below, y_{11} admittance parameter values is



Select one:

- ☐ a. $1/14$
- ☐ b. $-1/21$
- ☐ c. $2/7$
- ☐ d. $1/7$
- ☐ e. $4/7$

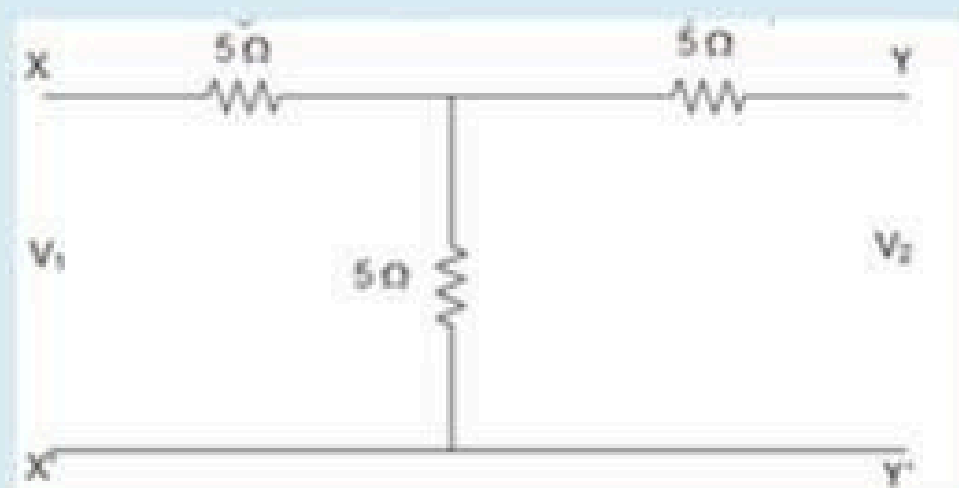
Question 9

Not yet answered

Marked out of 2.00

Flag question

For the circuit below, h_{21} hybrid parameter values is



Select one:

- ☐ a. 1
- ☐ b. 0.5
- ☐ c. 7.5
- ☐ d. 5
- ☐ e. 0.2

Question 10

Not yet answered

Marked out of 2.00

 Flag question

Let $V_{ab} = 300 \text{ V}$ in a balanced Y-connected three-phase generator with a phase sequence acb, then V_{bn} is:

Select one:

- ☐ a. $173\angle -30^\circ \text{ V}$
- ☐ b. $173\angle +30^\circ \text{ V}$
- ☐ c. $173\angle -150^\circ \text{ V}$
- ☐ d. $173\angle +150^\circ \text{ V}$
- ☐ e. $173\angle -270^\circ \text{ V}$

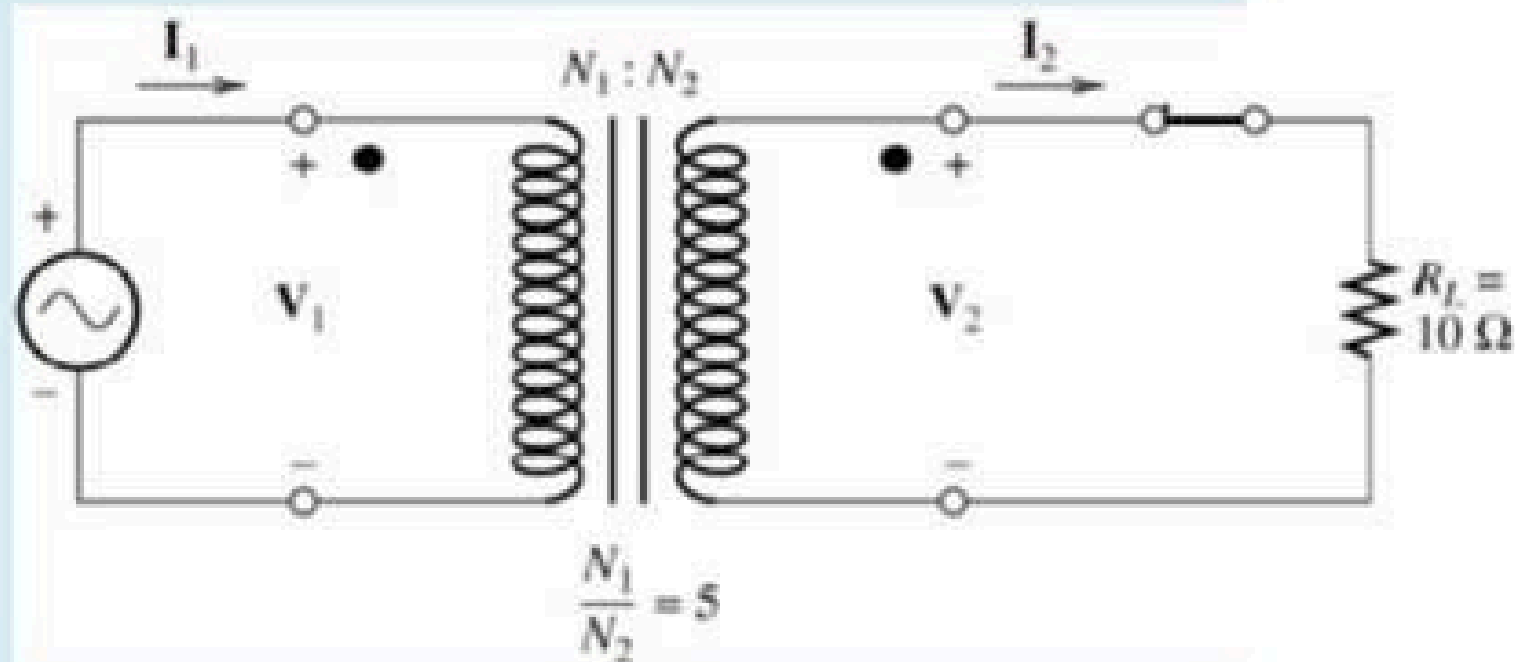
Question 11

Not yet answered

Marked out of 2.00

Flag question

If the voltage source in the figure is = 140 V, then I_1 is:



Select one:

- ☐ a. 0.44 A
- ☐ b. 0.56 A
- ☐ c. 0.4 A
- ☐ d. 0.52 A
- ☐ e. 0.48 A

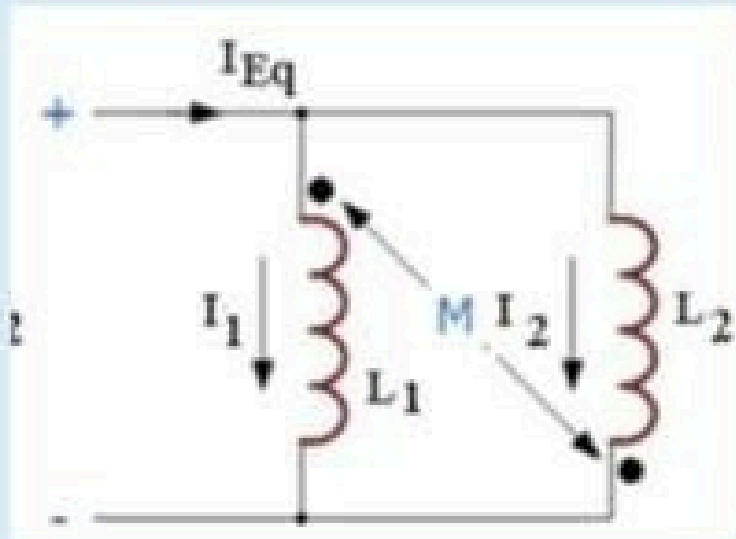
Question 12

Not yet answered

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Flag question

If $L_1 = L_2 = 6 \text{ mH}$ and $M = 1 \text{ mH}$, in the circuit the L_{Eq} is:



Select one:

- ☐ a. 0.5 mH
- ☐ b. 2 mH
- ☐ c. 1 mH
- ☐ d. 2.5 mH
- ☐ e. 1.5 mH

Question 13

Not yet answered

Marked out of 1.00

 Flag question

A transformer has a voltage 480 V and current 2.5 A on the primary side. And 80 V on the secondary. What is the secondary current on the load.

Select one:

- ☐ a. 30 A
- ☐ b. 20 A
- ☐ c. 10 A
- ☐ d. 5 A
- ☐ e. 15 A

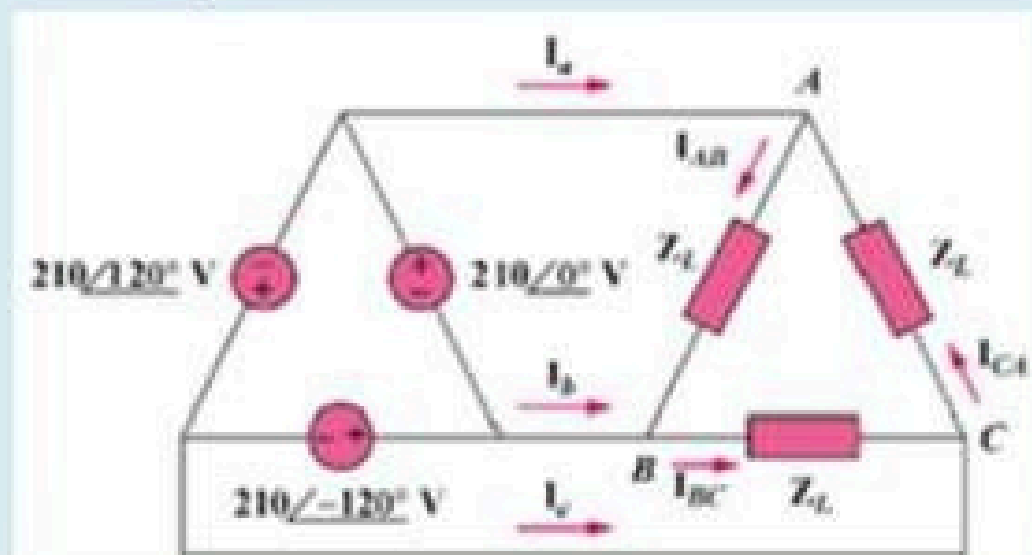
Question 14

Not yet answered

Marked out of 2.00

Flag question

The impedance $Z_L = 5 \angle 10^\circ$ Ohms in the circuit below, I_a is:



Select one:

- ☐ a. $72.75 \angle 80^\circ \text{ A}$
- ☐ b. $36.37 \angle -40^\circ \text{ A}$
- ☐ c. $72.75 \angle -40^\circ \text{ A}$
- ☐ d. $36.37 \angle -160^\circ \text{ A}$
- ☐ e. $72.75 \angle -160^\circ \text{ A}$

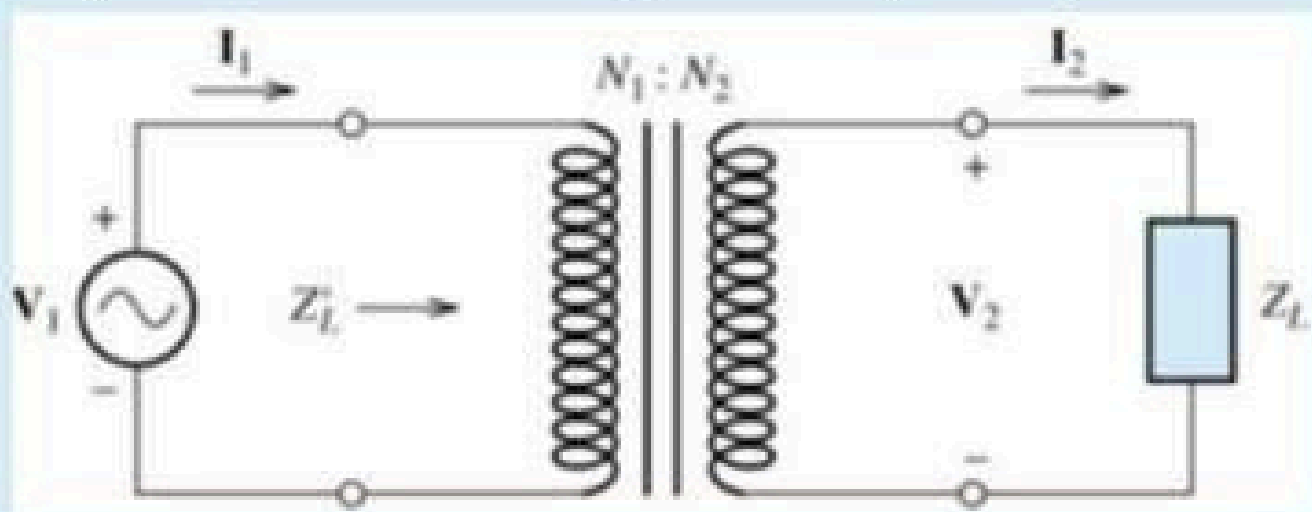
Question 15

Not yet answered

Marked out of 2.00

Flag question

Considering the figure below, let $N_1 = 100$, $N_2 = 10$, and impedance $Z_L = 1 \text{ k}\Omega$. Then the impedance seen looking into the primary is:



Select one:

- ☐ a. $10 \text{ k}\Omega$
- ☐ b. $1 \text{ k}\Omega$
- ☐ c. $10 \text{ M}\Omega$
- ☐ d. $100 \text{ k}\Omega$
- ☐ e. $1 \text{ M}\Omega$

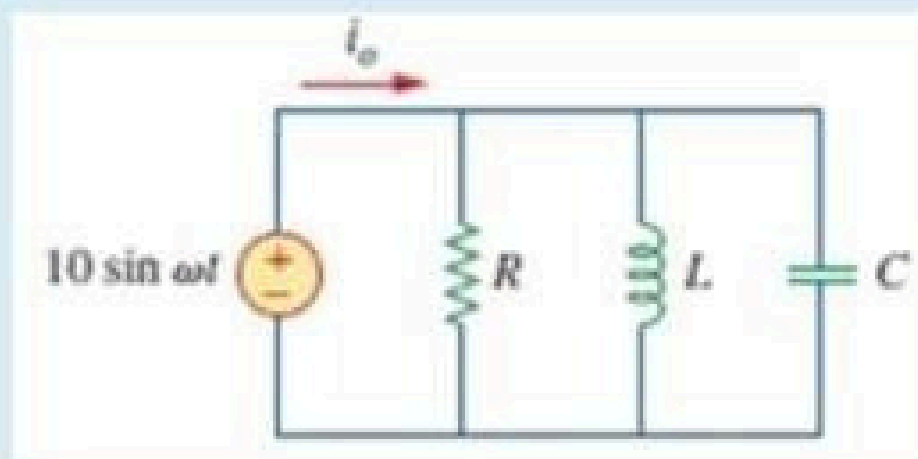
Question 16

Not yet answered

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Flag question

In the figure below, $R = 2\ \Omega$, $L = 1\ \text{mH}$, and $C = 0.4\ \mu\text{F}$. The average power dissipated in the circuit at resonance is:



Select one:

- ☐ a. 25 W
- ☐ b. 50 W
- ☐ c. 10 W
- ☐ d. 5 W
- ☐ e. 2 W

Question 17

Not yet answered

Marked out of 1.00

 Flag question

A series RLC with $R = 1 \, \Omega$, $L = 1 \, \text{mH}$, and $C = 0.4 \, \mu\text{F}$, the quality factor Q is:

Select one:

- ☐ a. 5
- ☐ b. 50
- ☐ c. 10
- ☐ d. 25
- ☐ e. 2

Question 18

Not yet answered

Marked out of 1.00

 Flag question

For high-pass active filter with corner frequency 20 kHz, then R and C are:

Select one:

- ☐ a. 10 k Ω , 7.9557 nF
- ☐ b. 2.5 k Ω , 7.9557 nF
- ☐ c. 1 k Ω , 7.9557 nF
- ☐ d. 5 k Ω , 7.9557 nF
- ☐ e. 2 k Ω , 7.9557 nF

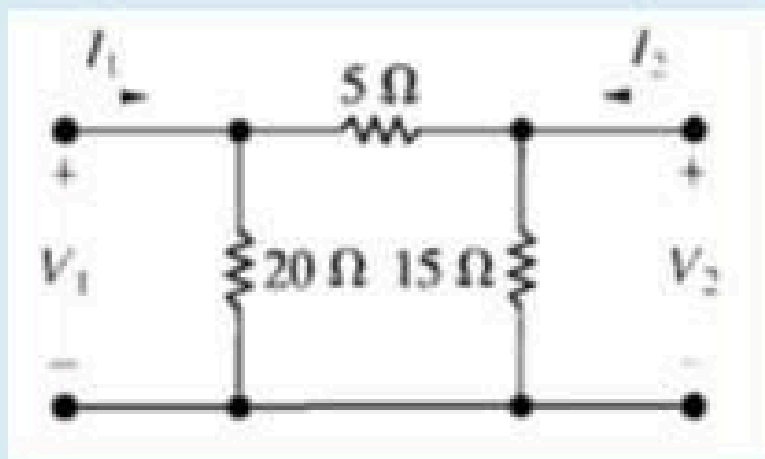
Question 19

Not yet answered

Marked out of 1.00

Flag question

For the circuit below, find z_{11} parameter.



Select one:

- ☐ a. $7.5\ \Omega$
- ☐ b. $20\ \Omega$
- ☐ c. $15\ \Omega$
- ☐ d. $10\ \Omega$
- ☐ e. $9.375\ \Omega$

Question 20

Not yet answered

Marked out of 2.00

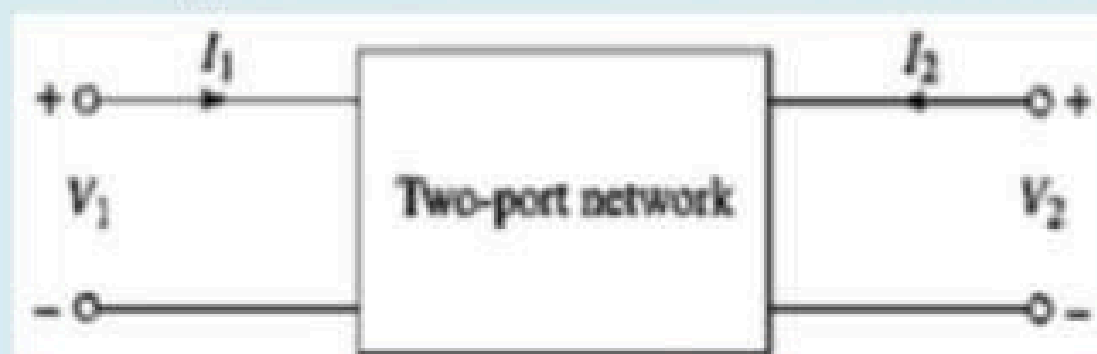
Flag question

For the network below, if

with the output port short-circuited: $V_1 = 25\text{ V}$,
 $I_1 = 1\text{ A}$, $I_2 = 2\text{ A}$

with the input port open-circuited: $V_1 = 10\text{ V}$, $V_2 = 50\text{ V}$, $I_2 = 2\text{ A}$

then h_{11} hybrid parameter values is:



Select one:

- ☐ a. 2
- ☐ b. 25
- ☐ c. 5
- ☐ d. 0.2
- ☐ e. 0.04

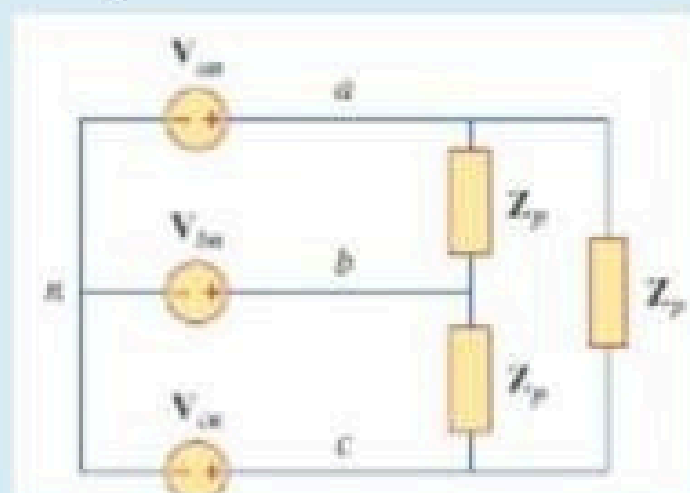
Question 21

Not yet answered

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Flag question

Given $V_{an} = 110 \angle 0^\circ$ V and impedance $Z_p = 3 + j4$ Ohms in the circuit below, Find the magnitude of the line current I_L



Select one:

- ☐ a. 41.57 A
- ☐ b. 38.11 A
- ☐ c. 34.64 A
- ☐ d. 48.5 A
- ☐ e. 45.03 A

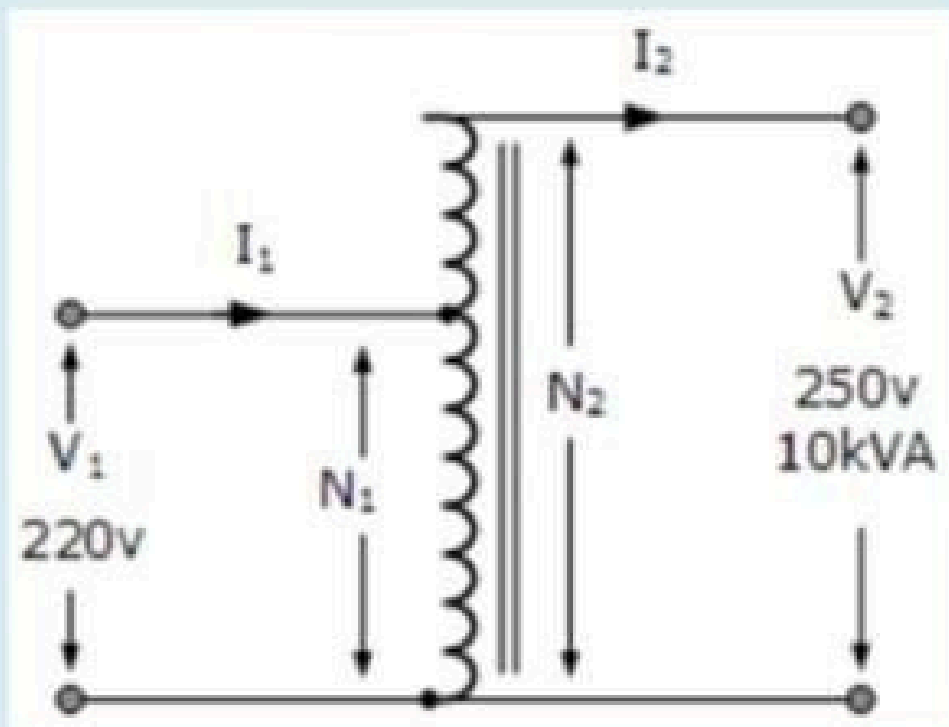
Question 22

Not yet answered

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Flag question

Considering the autotransformer shown in the figure, If the total coil turns in N2 side has 2400 turns, then N1 has:



Select one:

- ☒ a. 1760 turns
- ☐ b. 1848 turns
- ☐ c. 2112 turns
- ☐ d. 2024 turns
- ☐ e. 1936 turns

A three-phase system has an effective line voltage of 220 V, and is connected to a balanced delta load which draws 5 kW at a power factor of 0.8, then the line current is:

Select one:

- ☐ a. 12.03 A
- ☐ b. 12.89 A
- ☒ c. 15.04 A
- ☐ d. 16.4 A
- ☐ e. 13.88 A

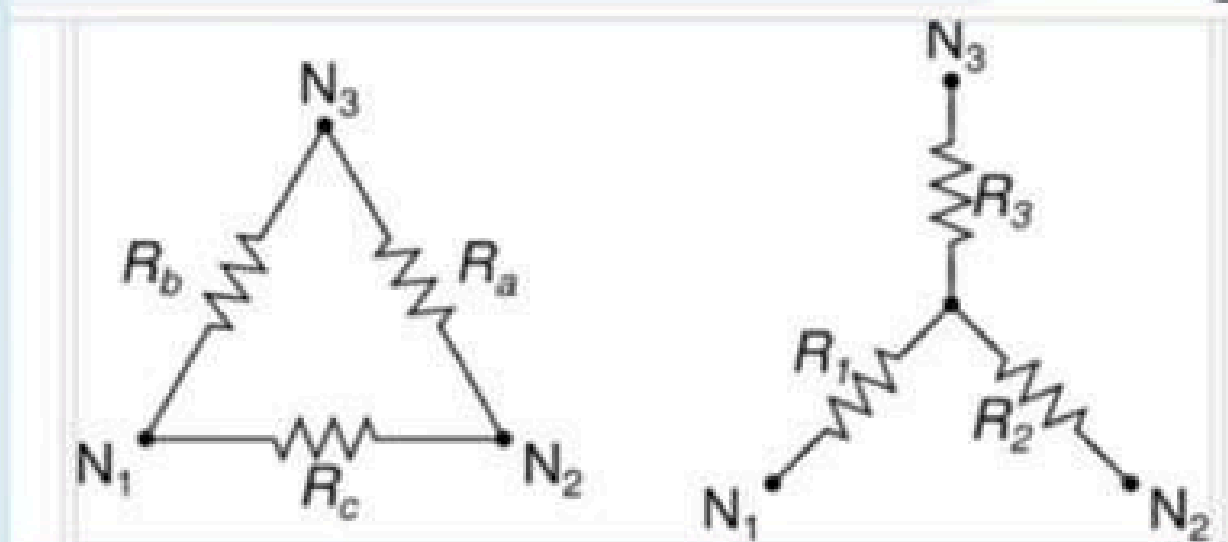
Question 25

Not yet answered

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Flag question

Consider that $R_a = R_b = R_c = 9\text{ Ohms}$ in the delta load in the figure, when converting to Y load R_3 becomes:



Select one:

- ☒ a. 3 Ohms
- ☐ b. 5 Ohms
- ☐ c. 4 Ohms
- ☐ d. 2 Ohms
- ☐ e. 1 Ohm